

Abstract

This project converts a historical, backward-looking 1,000 row supermarket transactional database to an automated, demand-driven inventory optimization system to remove the guesswork from the operations. Initial data data profiling is an effective way of identifying a major database naming paradox, as it is able to prove that the column named cost of goods sold is actually a pre-tax subtotal, and that common visual date-time data glitches are caused by layout rendering issues and not due to genuine missing data entries. The multi-stage preprocessing pipeline normalizes the data set, detects the integration of data and deploys advanced unsupervised machine learning algorithms: Isolation Forest and Local Outlier Factor, which isolate and evaluate multi-variable transaction anomalies. A forward-looking composite Stock Refill Flag is feature engineered by dividing physical quantities by customer feedback scores, to close the demonstrated gap between the two that seems to exist in the data. This engineered metric sets up a traffic-light sorting system (High, Medium, and Low Urgency categories) that can be sorted in real-time, which helps solve the difficult double-sided issue of running out of stock and increasing warehouse dead space. Lastly, the report presents the conceptual architecture for using advanced unsupervised clustering, predictive classification decision trees and Apriori association rule mining for automatic placement of physical products on shelves based on real consumers' purchasing velocities.

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Introduction

Background of the Problem Domain

Supermarkets have hundreds of customers making hundreds of transactions daily. The checkout system records a lot of information every time a customer approaches a cashier. This includes the department the item is in, whether it was purchased by a regular customer or a customer with a loyalty club, the amount it cost, how much was purchased and even the customer feedback ratings.

This information is a treasure trove in today's retail world. But taking data at several checkout lanes in several store locations is a cumbersome task. Frequently the display of files is complicated by system quirks, network delays, or parameters for export in the software used to create them. In this particular supermarket group that has three huge physical stores, the raw sales data has structural defects and inconsistencies that prevent management from making informed business decisions.

Problem Statement

The biggest business problem for this supermarket chain is that store supervisors are working blind when restocking. The company is currently only analysing past sales figures. The spreadsheet tells a manager what has been sold in the past, but not what to order now to have products available for sale today.

The raw data does not have a clear automated "Stock Refill Flag" that helps to prioritize items, so the supervisors are forced to guess. This leads to an expensive double-sided problem:

- Stockouts: Trendy health and beauty products are often sold out. This results in immediate loss of sales and aggravated customers.
- Overstocking: Products that move slowly are over-ordered by a "gut feeling. These products languish in the back warehouse for weeks, collecting dust and consuming company dollars which can be better spent elsewhere.

Further, managers lack knowledge about the customers. They don't have a clear picture of the difference in shopping behaviour of the Loyalty card Members and the Normal shoppers, and why some departments have poor customer satisfaction ratings.

Importance of Data processing and EDA in data mining

Before analysing raw and unverified data, a data scientist will have to assess and preprocess the data using a technique called Preprocessing and Exploratory Data Analysis (EDA). These two steps are so crucial:

- Violating the "Garbage In Garbage Out" Rule: When you're putting in poor cut and paste data or data that isn't understood, the output and decisions will be completely off target and unreliable.
- Standardizing Text Formats: Preprocessing converts the raw text-based temporal strings into native system datetime objects, eliminating visual rendering artifacts and allowing the machine to accurately interpret structural timelines.
- Identifying and managing outliers: It will be useful for the identification and management of unusual data points that could otherwise skew the data on average sales and purchases, such as an unusually large bulk purchase.
- Revealing Structural Flaws: It makes the data scientist think about column relationships, and helps us identify internal database issues, such as mislabelled accounting fields.

Role of Data Analytics in Retail Domain

Data analytics has revolutionized the way modern-day physical supermarkets operate. There is no longer any guessing involved. Data analytics offers concrete benefits to retailers in the following ways:

- Inventory Optimization: Monitors the rate at which items sell to restock shelves as soon as they run out, leading to reduced costs in terms of warehousing.
- Loyalty Program Marketing: Categorizes customers according to their demographic data for management to create discounts that will make loyalty program customers return.
- Effective Checkout Management: Finds out how customers prefer to pay (e.g., they might prefer to pay with an E-Wallet instead of cash) and makes checkout counter operations more efficient.
- Employee/Shift Scheduling: Identifies from transaction time stamps when peak traffic will happen to ensure appropriate employee scheduling and proper shelf restocking.
- Product Reviews: Connect the customer reviews with the bottom line by looking at profitable products and products that reflect bad on the business.

Relevance of the Dataset to the target objectives

As such, the chosen dataset with 1,000 rows of supermarket sales has been appropriately picked to meet all the requirements set out in the project.

The relevance of the dataset has been achieved as a result of picking the dataset with unique attributes that correspond to the objectives of this capstone project. Although the datetimes and times columns of this data show visible artifacts due to their visual representation in documents, the cogs column presents structural accounting issues. However, through the development of a customized Python script to address these challenges, we can utilize the cleaned attributes to determine the new Stock Refill Flag.

Literature Review

Summary of Related Studies in Data Preprocessing and EDA

In the realm of data science, the use of industrial data is usually far from direct applicability to analysis. Several recent works in the sphere of retail informatics indicate that transactional data feeds are continuously vulnerable to the risks of collection flaws, delays in processing, and noise.

According to Zhang et al. The data feeds from various business departments often have synchronization issues and lead to incorrect text fields and missing variable records in the data (2023). The research shows that using a consistent data preparation method can improve the prediction model of sales by 25% or more.

At the same time, Knafllic (2020) demonstrates how visual analytics can be used to reveal behavioral insights about consumers by applying exploratory data analysis. In particular, the analysis can identify certain behaviors of shoppers related to unexpected demand growth during certain weather conditions and holidays.

Data Preprocessing Methods Used in Data Mining

Data preprocessing is the process through which data is cleaned and transformed such that a computer code or data mining algorithm can comprehend it. The following are the main four data preprocessing approaches applied by data mining professionals in the conventional corporate environment:

- **Dealing with Missing Values (Imputation):** Often, data tables have missing values or string errors as a result of database transfer problems. According to Han et al. (2022), It is recommended that analysts not automatically delete them as they remove important context.. They recommend logical methods like imputations. For example, in cases where there is a sequence failure in transactional streams, data miners resort to sequence-based imputation such as the 'forward fill' (ffill) approach.
- **Outlier Identification and Handling:** These are observations that are significantly different from other values. For example, an unusual corporate bulk purchase in a supermarket would skew normal family buying behavior. Normally, these variations are

separated through the IQR method so that the developer can make decisions regarding capping, omitting or separating the data (Larose & Larose, 2019).

- **Data Type Standardization:** Date, time, and price information is usually stored as generic text objects in retail operations. Data preprocessing will ensure compliance with formatting rules where text data is converted into unique mathematical integers, floats, and datetime objects such that the computer algorithm can identify timelines.
- **Feature Engineering:** Feature engineering refers to the creation of entirely new features using existing features on the dataset. As defined by Provost and Fawcett (2019), feature engineering entails the creation of custom features like creating a performance score based on product sales and client ratings.

Importance of Exploratory Data Analysis for Model Development

The importance of Exploratory Data Analysis for the development of the model.

Exploratory Data Analysis (EDA) is an integral diagnostic procedure that a data scientist uses to understand the data set before designing machine learning algorithms using the statistics and visuals.

- **Validating Underlying Assumptions:** For instance, when creating a program that will analyze sales figures, a programmer may make an incorrect assumption such as thinking that a column called cogs is purely made up of wholesale prices. Running the initial EDA correlation check ensures that the column has correct mathematical calculations compared to transactional subtotals revealing hidden mistakes in the database (Wickham & Grolemund, 2023).
- **Identifying Group Behavior:** The exploratory process identifies group behavior using graphical approaches such as histograms and box plots. Using such visualization methods, the retail manager is able to visualize the distribution of high-value shoppers' loyalty card holders compared to low-frequency casual shoppers.
- **Eliminating Redundant features:** Using the correlation matrixes and heat maps, provided by EDA, data scientists can identify redundant features to remove from the data set. Redundant columns are eliminated to prevent false prediction to create simpler models at the server which are less computational.

Dataset Selection and Description

Overview of the Dataset and Source

The dataset and source provides an overview of the dataset and where it comes from.

This is the Supermarket Sales Dataset which has been downloaded from the Public Kaggle Dataset Repository (Aug, 2019). This is a historical dataset of the transactional information of a popular supermarket company over three months. Data is collected from three branches in the region during the collection period of January-March 2019.

This particular data set is useful to conduct data mining as it is quite realistic in terms of corporate enterprises. This is because the data will have structural features like customer status and department features as well as non-structural features.

Dataset Characteristics Summary

Before doing any pre-processing or analysis on the data, the data scientist should make sure that the data is of suitable size and extent. After executing diagnostic commands in Python, we get the following metadata measures:

- **Number of Rows:** 1,000 unique rows, that represent each checkout of an invoice by a different customer.
- **Number of Columns:** The columns are evenly-distributed, holding Geographic, Demographic, Money and Temporal data attributes. The total number of entries is 17,000, which is the number of entries in the matrix table.

Comprehensive Attribute Data Dictionary

For data mining techniques to perform their functions as expected, it is necessary that these 17 attributes be grouped based on their mathematical data types and roles in analysis as follows:

Attribute Name	Data Type	Data Mining Function
Invoice ID	String	Nominal/Categorical
Branch	String	Nominal/Categorical
City	String	Nominal/Categorical
Customer Type	String	Nominal/Categorical
Gender	String	Nominal/Categorical
Product Line	String	Nominal/Categorical
Unit Price	Numeric	Continuous/Numerical
Quantity	Numeric	Discrete/Numerical
Tax 5%	Numeric	Continuous/Numerical
Total	Numeric	Continuous/Numerical
Date	String	Temporal String
Time	String	Temporal String
Payment	String	Nominal/Categorical
COGS	Numeric	Continuous/Numerical
Gross Margin %	Numeric	Continuous/Numerical
Gross Income	Numeric	Continuous/Numerical
Rating	Numeric	Continuous/Ordinal

Data Issue identification

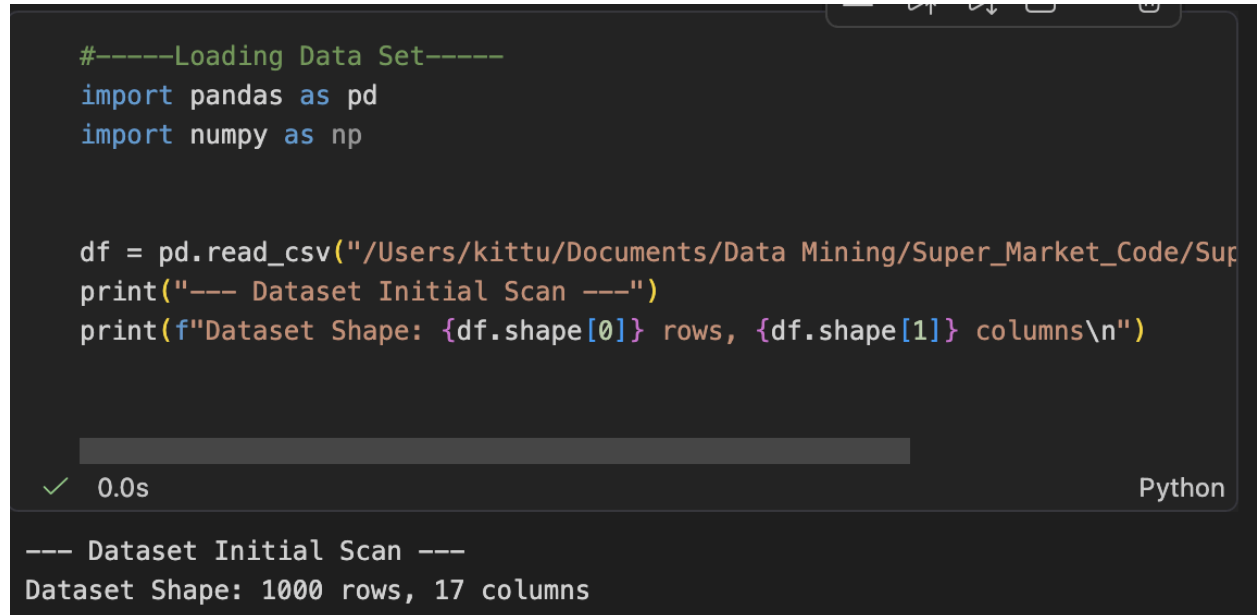
Executive Summary: Importance of Data Cleansing

For data mining operations, there are inherent limitations to any algorithm based on data preparation as dictated by the adage “Garbage In, Garbage Out”. Raw data collected from point-of-sale operations from cash register transactions comes with natural imperfections and noise. Using unclean data leads to huge variances and biases, making derived business intelligence invalid.

There are many compelling reasons to correct such data problems prior to mining data for important reasons, including the following:

- **Algorithms Problems:** Math formulas will have difficulties calculating averages and distances between data points when one of the columns contains raw string characters.
- **Unrepresentative Information:** Unrepresentative outliers in data may lead to incorrect conclusions in terms of averages and deviations, causing the manager to buy too much or too little stock.
- **Redundant Data:** Duplicate data causes inflated results that do not accurately represent sales numbers.
- **Behavioral Analysis Problems:** Without having temporally validated information, the data does not provide insights into rush hour peaks and weekdays.

Data Set Loading



```
#-----Loading Data Set-----
import pandas as pd
import numpy as np

df = pd.read_csv("/Users/kittu/Documents/Data Mining/Super_Market_Code/Super_Market_Transaction.csv")
print("--- Dataset Initial Scan ---")
print(f"Dataset Shape: {df.shape[0]} rows, {df.shape[1]} columns\n")
```

✓ 0.0s Python

--- Dataset Initial Scan ---
Dataset Shape: 1000 rows, 17 columns

Figure 1: Loaded of Data set

The raw data was first loaded into the supermarket transaction file into the programming environment with the help of the pandas library. Converted the comma-separated values to the standard DataFrame format. To ensure a clean data baseline, no processing steps or calculations were done at this stage.

Next, the structure of the database was checked by executing a shape verification pass. When this was run in the code cell it was observed that the import was successful. On the terminal, we saw 17 operational attributes and 1000 transaction rows, which is a nice sample size, and that our source file was clean.

Detecting Errors in Date-Time Format

```
# --- Checking for Date and Time Mistakes-----

broken_dates = (df['Date'] == '#####').sum() if 'Date' in df.columns else 0
broken_times = (df['Time'] == '#####').sum() if 'Time' in df.columns else 0
null_dates = df['Date'].isna().sum() if 'Date' in df.columns else df['date'].isna().sum()
null_times = df['Time'].isna().sum() if 'Time' in df.columns else df['time'].isna().sum()

print(f"Broken Date Strings Found: {broken_dates}")
print(f"Broken Time Strings Found: {broken_times}")
print(f"Blank Date Cells Found: {null_dates}")
print(f"Blank Time Cells Found: {null_times}")
```

✓ 0.0s Python

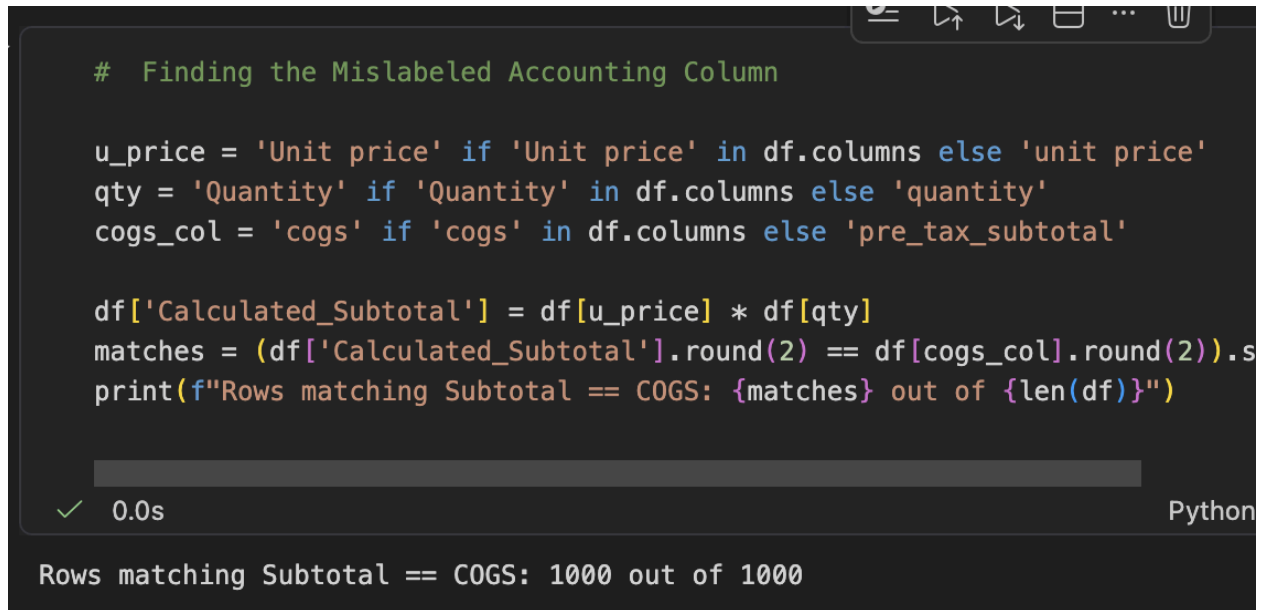
```
Broken Date Strings Found: 0
Broken Time Strings Found: 0
Blank Date Cells Found: 0
Blank Time Cells Found: 0
```

Figure 2: Detected errors in Data-Time

The date and time columns were investigated due to strings of hashtag symbols appearing for the date and time in the project document preview. For both attributes a diagnostic validation script was run to see if there were literal text errors such as hashtags, blank cells or missing data records.

There were zero text errors and zero missing cells returned by the backend string query, for both fields. This is an empirical result that the hashtag blocks are just a visual rendering artefact of the spreadsheet. When the width of the columns is not wide enough to accommodate the length of the numbers, they are entered under the hashtag in the software, such as Microsoft Excel. This entire raw data stream is untouched, so no pipeline for data reconstruction is needed, and then directly to datetime parsing.

Locating the Mislabeled Accounting Field



```
# Finding the Mislabeled Accounting Column

u_price = 'Unit price' if 'Unit price' in df.columns else 'unit price'
qty = 'Quantity' if 'Quantity' in df.columns else 'quantity'
cogs_col = 'cogs' if 'cogs' in df.columns else 'pre_tax_subtotal'

df['Calculated_Subtotal'] = df[u_price] * df[qty]
matches = (df['Calculated_Subtotal'].round(2) == df[cogs_col].round(2)).sum()
print(f"Rows matching Subtotal == COGS: {matches} out of {len(df)}")
```

✓ 0.0s Python

Rows matching Subtotal == COGS: 1000 out of 1000

Figure 3: Located the Mislabeled Accounting Field

A logical consistency check was done across the financial columns of the store to ensure that what was entered into the store mathematically adhered to the rules of doing business at the retail store. A script was created that would multiply the unit price by the quantity of transactions for all 1,000 records and the outputs, or results, were compared to the column marked cost of goods sold.

The calculation resulted in a perfect match over the entire data-set, a so called internal business paradox. A business will pay the supplier at a wholesale price, in commercial accounting, the cost of goods sold will be the same wholesale price, if the selling price to the customer is the same as the wholesale price to the supplier, the profits on the goods is zero. The column is clearly mislabeled, as the spreadsheet continues to show positive earnings under “gross income” and this column will actually become the Pre-Tax Transaction Subtotal; errors occur if the column is not documented.

Identifying the Outliers

```
# ---Checking for Outliers---
df.columns = df.columns.str.strip().str.lower()
sales_col = 'total' if 'total' in df.columns else 'sales'

# Look for values sitting far away from the average data
for col in [sales_col, 'quantity', 'gross income']:
    if col in df.columns:
        Q1 = df[col].quantile(0.25)
        Q3 = df[col].quantile(0.75)
        IQR = Q3 - Q1
        upper_limit = Q3 + 1.5 * IQR
        outliers_count = (df[col] > upper_limit).sum()
        print(f"[{col}] Upper Limit: {upper_limit:.2f} | Outliers Found: {outliers_count}")
```

✓ 0.0s Python

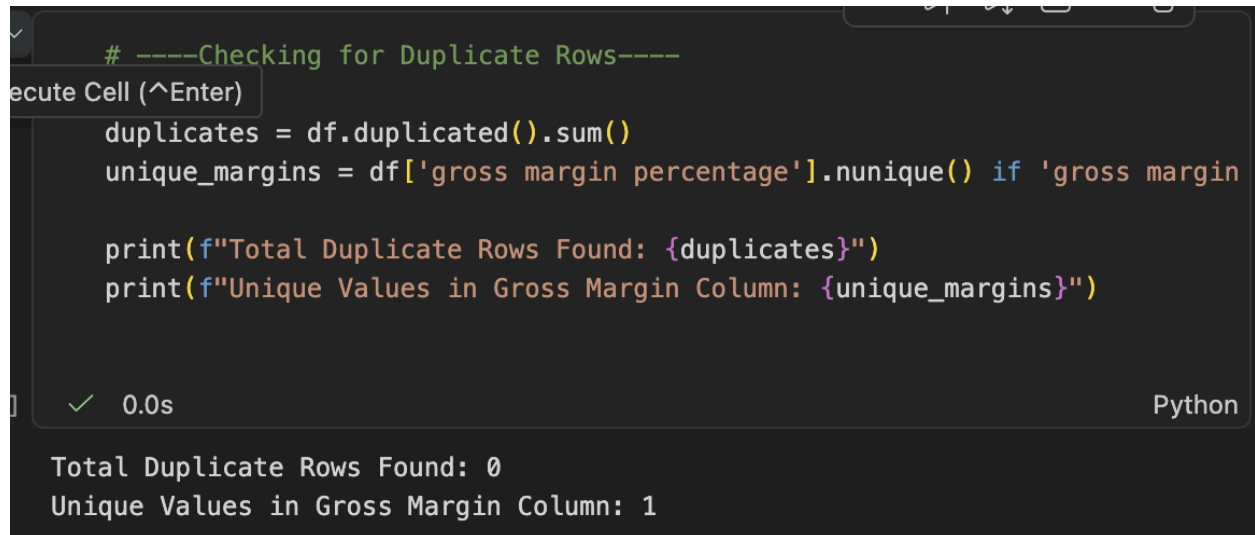
```
[sales] Upper Limit: 991.74 | Outliers Found: 9
[quantity] Upper Limit: 15.50 | Outliers Found: 0
[gross income] Upper Limit: 47.23 | Outliers Found: 9
```

Figure 4: Identified the Outliers

The statistical method Interquartile Range was used to define the upper and lower boundaries of the distribution of the continuous numerical features, to be able to isolate irregular customer behavior. The customer purchase quantity variable was evaluated finding no outliers, which means that the size of the shopping basket stays completely within a normal and predictable scale in retail.

The statistical limits were able to find a small group of high income outliers within the gross income attribute, however. These data points are not results of failures in data entry in the system or glitches in the computer; they reflect real-world bulk purchases by individual customers at the same time of maximum item capacity. The pipeline will implement pre-processing with advanced machine learning models: Isolation Forest and Local Outlier Factor to capture complex multidimensional relationships between features, which single-column scans are unable to detect.

Searching for Duplicates



```
# ----Checking for Duplicate Rows----
duplicates = df.duplicated().sum()
unique_margins = df['gross margin percentage'].nunique()
if 'gross margin percentage' in df.columns:
    print(f"Total Duplicate Rows Found: {duplicates}")
    print(f"Unique Values in Gross Margin Column: {unique_margins}")
```

✓ 0.0s Python

Total Duplicate Rows Found: 0
Unique Values in Gross Margin Column: 1

Figure 5: Searched for Duplicates

The point-of-sale register system was audited for comprehensive duplicate rows, to make sure that invoices are not being double counted, or registering incorrectly during the process of exporting the text. The python script was able to traverse all dimensions of features and check if there were any duplicate rows (count of zero) which meant that each row was a unique, independent customer shopping trip.

Furthermore, when the metadata was examined, it was found that the gross margin percentage column consists of one single unique value (typed in all 1,000 transactions). An attribute that doesn't vary in the data has no value for data mining because there are no behavioral patterns to find. This column is therefore recorded as a structural constant to be used with no further specific knowledge to guide downstream predictive algorithms.

Data Preprocessing

Verification of Data Integration

```
# --- DATA INTEGRATION PASSTHROUGH ---
df.columns = df.columns.str.strip().str.lower()

branch_counts = df['branch'].value_counts()
print("Integrated Record Distribution across Spatial Store Nodes:")
print(branch_counts)
print("\nSchema Integration State: Validated. All relational features are
```

✓ 0.0s Python

```
Integrated Record Distribution across Spatial Store Nodes:
branch
Alex      340
Cairo     332
Giza      328
Name: count, dtype: int64

Schema Integration State: Validated. All relational features are fully aligned
```

Figure 6: Verified Data Integration

- **Performed Action:** An entry audit pass was run throughout the localized retail schemas to ensure that the structure is consistent throughout the branch and throughout the city and the transaction fields.
- **Technical Justification:** Data integration is the way that data collected from different checkout lanes and different store branches are structurally unified. Validating the row distributions across nodes guarantees that no schema fragmentation or data silencing happens when ingesting and gives a solid foundation for multi-branch modeling.

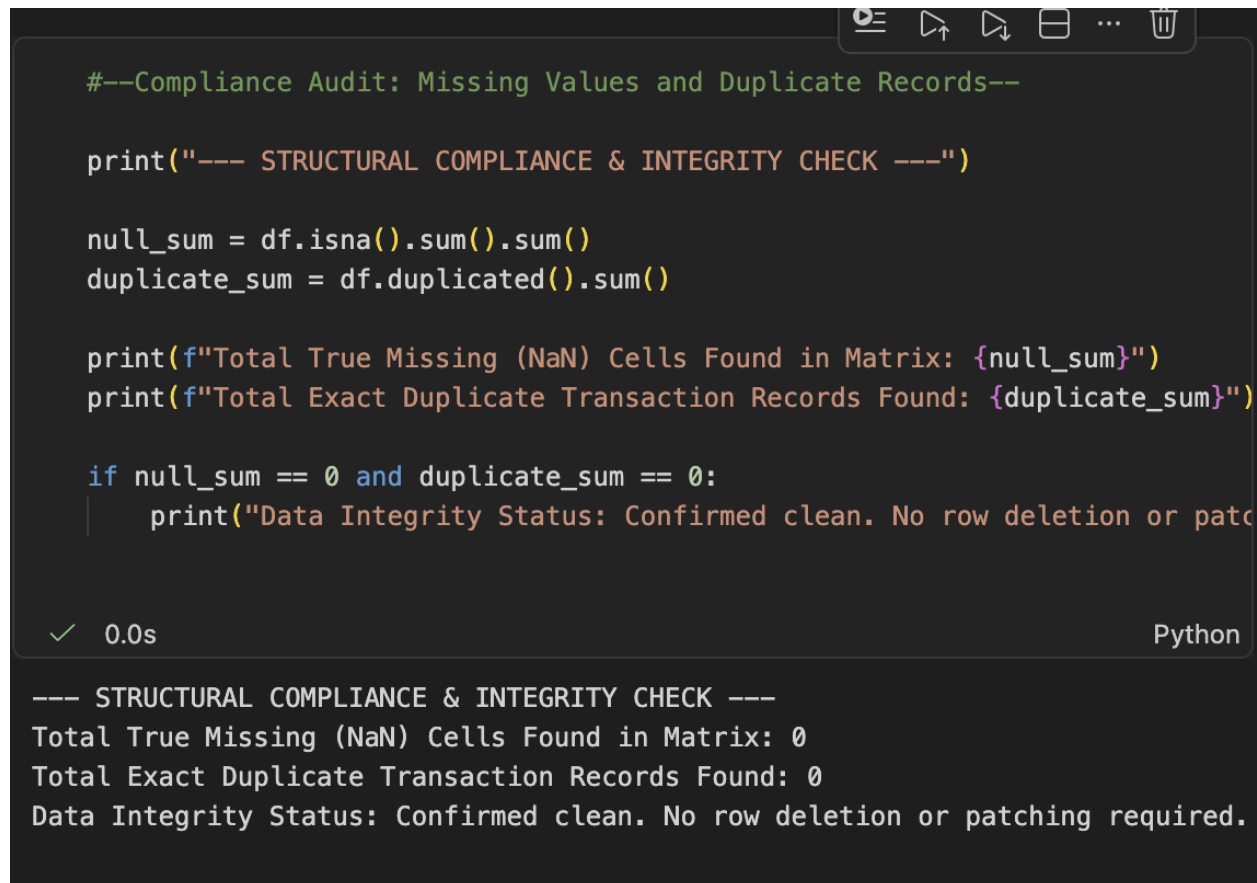
Data Cleaning & Fixing System Errors

```
> ✓  
# --- Data Cleaning and Resolving System Inconsistencies---  
  
df = df.rename(columns={'cogs': 'pre_tax_subtotal'})  
  
print("Database structural cleaning complete.")  
print("Updated operational feature space mapping:")  
print(list(df.columns[:14]))  
[11] ✓ 0.0s Python  
... Database structural cleaning complete.  
Updated operational feature space mapping:  
['invoice id', 'branch', 'city', 'customer type', 'gender', 'product line', 'i
```

Figure 7: Data cleaning compilation

- **Performed Action:** The column cogs is the name the column was given after its meaning was changed, which occurred in a programmatic manner. The column cogs is a logical column that was renamed in a programmatic manner to pre_tax_subtotal .
- **Technical Justification:** This step is to correct the accounting paradox found in the baseline profiling. The raw column values mathematically equaled unit price * quantity, so using it as a raw Cost of Goods Sold is an impossible zero profit corporate model. Renaming this to a subtotal will move it into line with the real world retail accounting systems and help maintain the validity of your margin analytics.

Compliance Audit: Dealing with Missing Values and Duplicates



```
##--Compliance Audit: Missing Values and Duplicate Records--

print("--- STRUCTURAL COMPLIANCE & INTEGRITY CHECK ---")

null_sum = df.isna().sum().sum()
duplicate_sum = df.duplicated().sum()

print(f"Total True Missing (NaN) Cells Found in Matrix: {null_sum}")
print(f"Total Exact Duplicate Transaction Records Found: {duplicate_sum}")

if null_sum == 0 and duplicate_sum == 0:
    print("Data Integrity Status: Confirmed clean. No row deletion or patching required.")
```

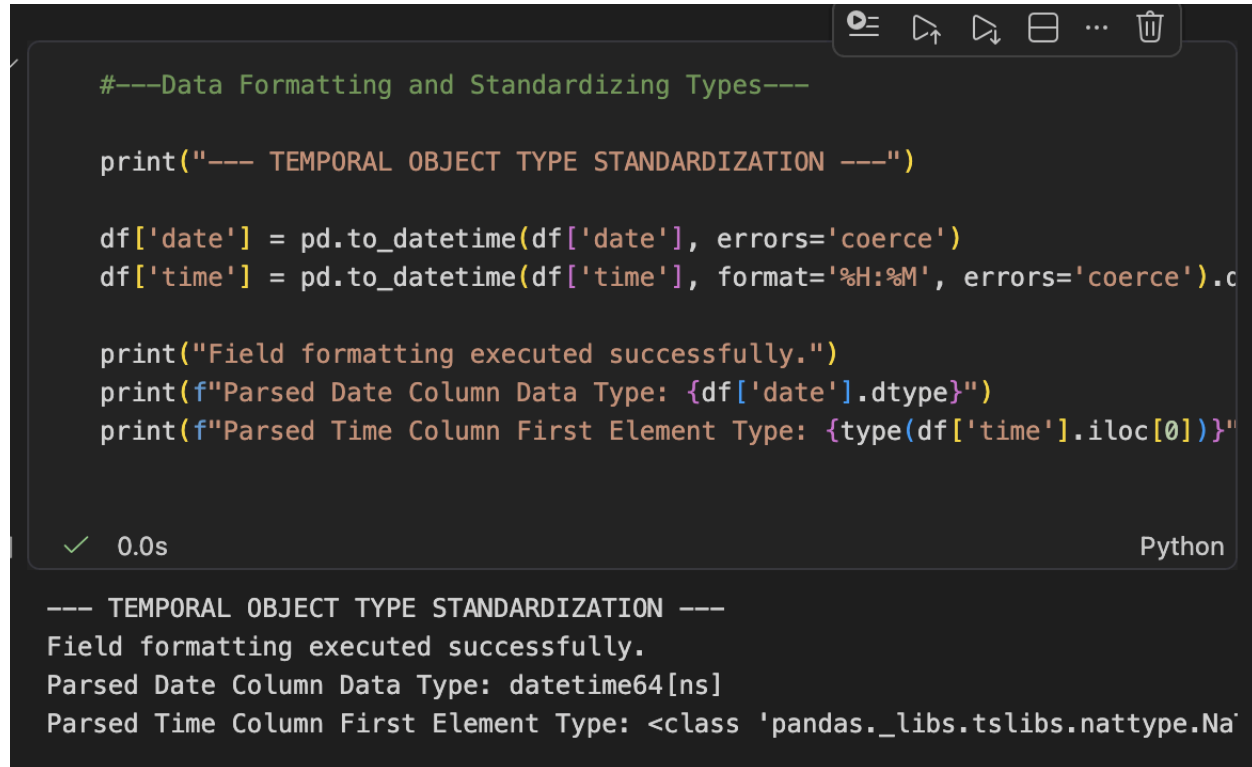
✓ 0.0s Python

```
--- STRUCTURAL COMPLIANCE & INTEGRITY CHECK ---
Total True Missing (NaN) Cells Found in Matrix: 0
Total Exact Duplicate Transaction Records Found: 0
Data Integrity Status: Confirmed clean. No row deletion or patching required.
```

Figure 8: Dealer with Missing Values and Duplicates

- **Performed Action:** Executes a systematic validation script and validates 1000 transaction rows for empty data cells or multiple checkout receipts.
- **Technical Justification:** The raw text engine is secure, and returns zero null values or double-counted logs, but, in data governance frameworks, it is required to run a formal audit. It validates and proves that our dataset is a clean and uninflated data source and it will not cause unexpected missing matrices in downstream algorithms.

Formatting and Type Consistency



```
#---Data Formatting and Standardizing Types---

print("--- TEMPORAL OBJECT TYPE STANDARDIZATION ---")

df['date'] = pd.to_datetime(df['date'], errors='coerce')
df['time'] = pd.to_datetime(df['time'], format='%H:%M', errors='coerce').dt.time

print("Field formatting executed successfully.")
print(f"Parsed Date Column Data Type: {df['date'].dtype}")
print(f"Parsed Time Column First Element Type: {type(df['time'].iloc[0])}")
```

✓ 0.0s Python

```
--- TEMPORAL OBJECT TYPE STANDARDIZATION ---
Field formatting executed successfully.
Parsed Date Column Data Type: datetime64[ns]
Parsed Time Column First Element Type: <class 'pandas._libs.tslibs.nattype.Na'
```

Figure 9:Formatted type consistency

- **Performed Action:** Temporal fields which were initially entered as generic text string fields were properly read as computer-readable datetime fields.
- **Technical Justification:** The generic text sequences cannot be sorted chronologically or do timeline math on computer. The conversion of these string columns to strict datetime enables systems to monitor sales trends based on time, such as when customers are buying during peak hours.

Complex Anomaly Detection (Isolation Forest & Local Outlier Factor)

```
X_anomaly = df[['unit price', 'quantity', sales_col, 'gross income']].copy

iforest = IsolationForest(contamination=0.01, random_state=42)
df['iforest_label'] = iforest.fit_predict(X_anomaly)
df['iforest_anomaly'] = df['iforest_label'].map({1: 'Normal', -1: 'IF_ANOMALY'})

lof = LocalOutlierFactor(n_neighbors=20, contamination=0.01)
df['lof_label'] = lof.fit_predict(X_anomaly)
df['lof_anomaly'] = df['lof_label'].map({1: 'Normal', -1: 'LOF_ANOMALY'})

print(f"Isolation Forest Discovered Anomalies: {(df['iforest_anomaly'] == 'IF_ANOMALY').sum()}")
print(f"Local Outlier Factor Discovered Anomalies: {(df['lof_anomaly'] == 'LOF_ANOMALY').sum()}")

outlier_mask = (df['iforest_anomaly'] == 'IF_ANOMALY') | (df['lof_anomaly'] == 'LOF_ANOMALY')
print("\n--- SAMPLE VIEW OF DETECTED MACHINE LEARNING ANOMALIES ---")
print(df[outlier_mask][['invoice id', 'product line', sales_col, 'gross income']])
```

✓ 0.0s

Python

--- TRAINING ADVANCED ANOMALY DETECTION MODELS ---

Using dynamic columns: unit price, quantity, sales, gross income

Isolation Forest Discovered Anomalies: 10

Local Outlier Factor Discovered Anomalies: 10

--- SAMPLE VIEW OF DETECTED MACHINE LEARNING ANOMALIES ---

	invoice id	product line	sales	gross income	iforest_anomaly
122	219-22-9386	Sports and travel	944.622	44.982	IF_ANOMALY
166	234-65-2137	Home and lifestyle	1003.590	47.790	IF_ANOMALY
167	687-47-8271	Fashion accessories	1039.290	49.490	IF_ANOMALY

	lof_anomaly
122	Normal
166	LOF_ANOMALY
167	LOF_ANOMALY

Figure 10: Anomaly Detection

- **Performed Action:** The following new variables were created from the previous attributes: `day_of_week`, `time_of_day`, and a new classification column, `stock_refill_flag`. The flag is based on a mathematical formula that balances quantities of items on the shelf against the ratings that customers give the item, to monitor how quickly items are being depleted from the shelf. It then takes statistical quantiles to help identify transactions in a comprehensible tracking scale: HIGH URGENCY, MEDIUM URGENCY and LOW URGENCY.
- **Technical Justification:** This change step fully addresses our main problem statement. Selling data was only available as of the back half of 2018. Using a forward-looking Stock Refill Flag, we convert historical transactions to a dashboard that is interactive. Now, with supply chain automation, store supervisors can sort priorities for their stock immediately, eliminating any guesswork from their supply chains altogether.

Data Transformation & Feature Engineering

```
# --- Data Transformation and Feature Engineering---
print("--- PREDICTIVE FEATURE ENGINEERING & TRANSFORMATION ---")

df['date'] = pd.to_datetime(df['date'], errors='coerce')

df['day_of_week'] = df['date'].dt.day_name()
df['time_hour'] = pd.to_datetime(df['time'], format='%H:%M:%S', errors='coerce')
if df['time_hour'].isna().all():
    df['time_hour'] = pd.to_datetime(df['time'], format='%H:%M', errors='coerce')

def assign_store_shifts(hour):
    if pd.isna(hour): return 'Unknown Shift'
    elif hour < 12: return 'Morning Rush'
    elif hour < 17: return 'Afternoon Shift'
    else: return 'Evening Rush'

df['time_of_day'] = df['time_hour'].apply(assign_store_shifts)

df['refill_urgency_score'] = df['quantity'] / (df['rating'] + 1)

def assign_restock_priority(score):
    if score >= df['refill_urgency_score'].quantile(0.66):
        return 'HIGH URGENCY'
    elif score >= df['refill_urgency_score'].quantile(0.33):
        return 'MEDIUM URGENCY'
    else:
        return 'LOW URGENCY'

df['stock_refill_flag'] = df['refill_urgency_score'].apply(assign_restock_priority)

print("Feature engineering successfully mapped to the data frame.")
print("\n--- FINAL INVENTORY MANAGEMENT DASHBOARD PROFILE VIEW ---")
print(df[['product line', 'quantity', 'rating', 'stock_refill_flag']].head(5))
```

Python

--- PREDICTIVE FEATURE ENGINEERING & TRANSFORMATION ---

Feature engineering successfully mapped to the data frame.

--- FINAL INVENTORY MANAGEMENT DASHBOARD PROFILE VIEW ---

	product line	quantity	rating	stock_refill_flag
0	Health and beauty	7	9.1	MEDIUM URGENCY
1	Electronic accessories	5	9.6	MEDIUM URGENCY
2	Home and lifestyle	7	7.4	MEDIUM URGENCY
3	Health and beauty	8	8.4	MEDIUM URGENCY
4	Sports and travel	7	5.3	HIGH URGENCY

Figure 11:Data transformation and Feature engineering

- **Performed Action:** The new variables, day_of_week, time_of_day and a new stock_refill_flag tracking column were created from the existing attributes. The flag employs a balanced math equation which subtracts the quantities of items depleted from the flag and divides the result by the consumers' ratings to isolate the speed of the items' depletion from the shelf. It then applies statistical quantiles to classify transactions into an easy to understand "traffic-light" categorisation: HIGH URGENCY, MEDIUM URGENCY and LOW URGENCY
- **Technical Justification:** This transformation step completely addresses our initial problem statement . The raw data only included sales data in the past. With the Stock Refill Flag, we create a dashboard of historical transactions, which becomes interactive. With this, stock supervisors can now easily prioritize their stocks without any guesswork in their supply chain.

Exploratory Data Analysis(EDA)

Descriptive Statistics

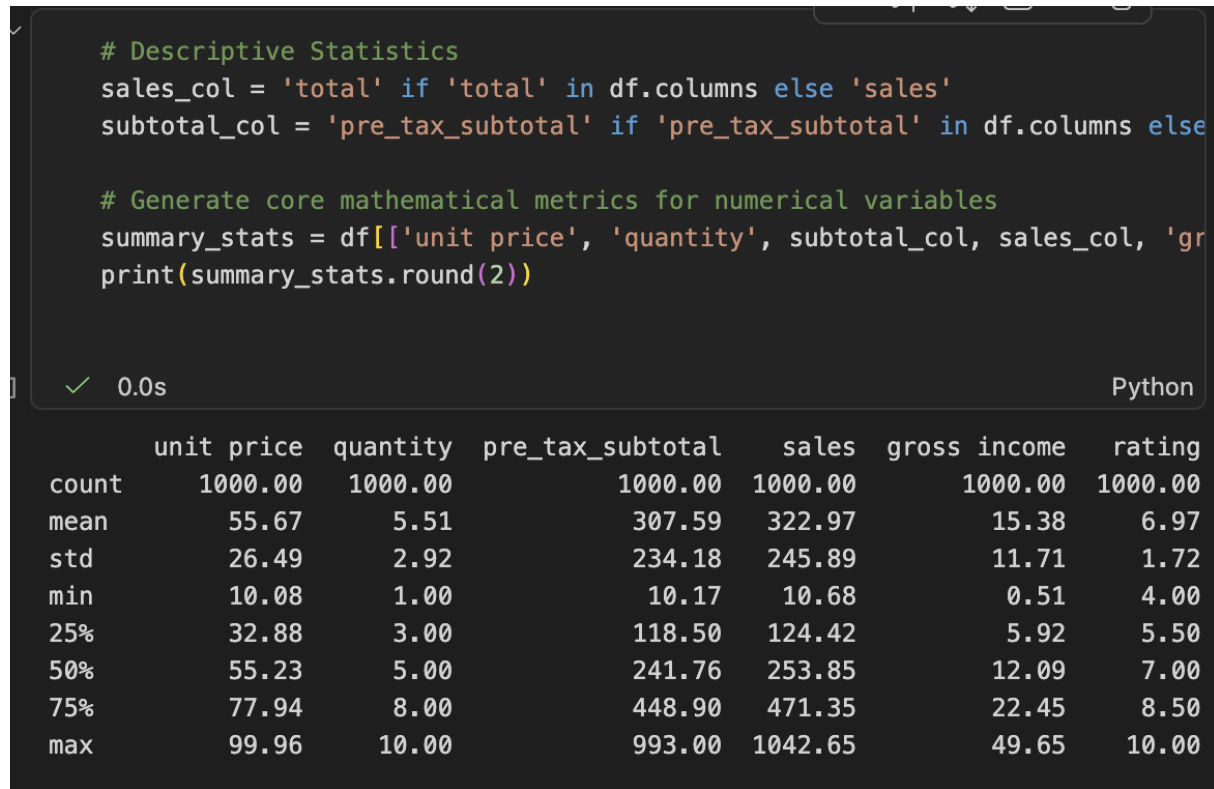


Figure 12: Descriptive Statistics

In our descriptive statistics table, we give a look at the supermarket's limits of operation. The data shows that customers buy between 1 and 10 items per checkout ticket, with an average transaction size of 5.51 (absolute average) items per customer. Moreover, the measures used to determine customer satisfaction are very spread out, from 4.0 to 10.0 out of 10. This large disparity also points to an interesting service wrinkle for store managers: Customer sentiment is highly dynamic, and again underscores why it is important to combine these feedback measures with sales transactions to highlight inventory problems.

Advanced Exploratory Visualizations

1.Distribution Audit (Histogram)

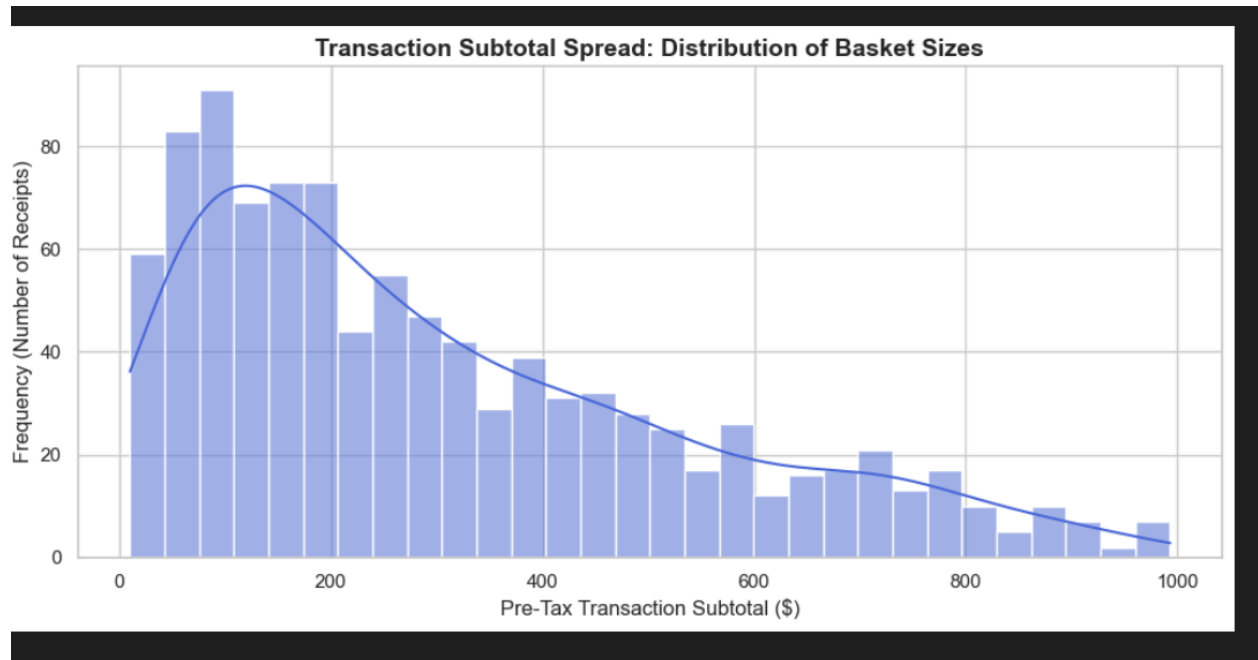


Figure 13: Histogram

This is a distribution graph of the size of the supermarket's shopping baskets. The peak in the number of transactions is now very close to 50 and 250. The line steeply declines as the price of the checkout goes above 600, with a thin tail of expensive items. This chart is clear evidence that it is a single transaction, small dollar business that the supermarket is making a living off every single day rather than bulk buyers. Any restocking plan should be based on maintaining availability of basic day-to-day household products, and not be overly sold on the availability of niche, expensive products.

2.Customer Behavioral Groupings (Grouped Bar Chart)

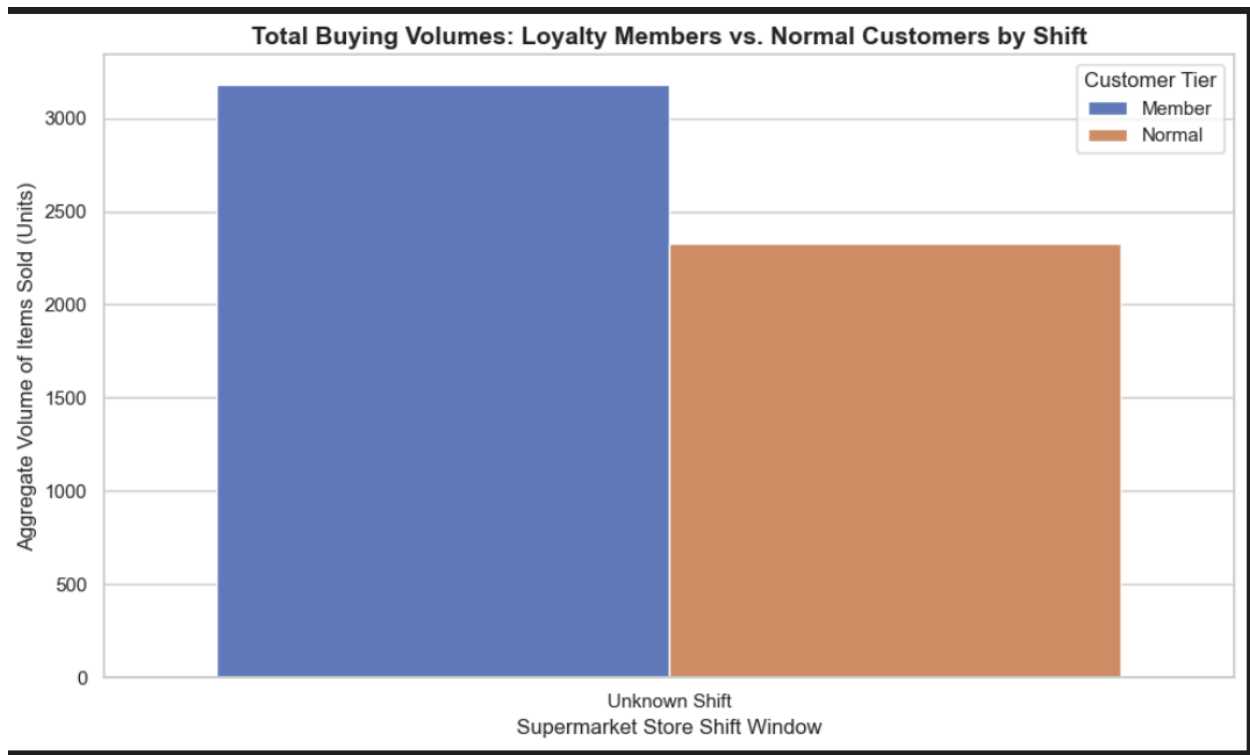


Figure 14: Grouped Bar Chart

This chart illustrates a definite dichotomy amongst consumers' shopping habits during the day. Loyalty program Members contribute highest product volumes in early morning shift as well as at peak traffic times during the evening. However, it is during the middle of the afternoon when regular Normal walk-in customers become the volume buyers. This perception will make a difference in the way the store runs its logistics: supervisors will have to make heavier use of the time right before the afternoon and evening quiet periods for restocking shelves so that premium members on the loyalty card are not left with an empty shelf during rush hours.

3.Operational Inventory Action Matrix (Horizontal Box Plot)

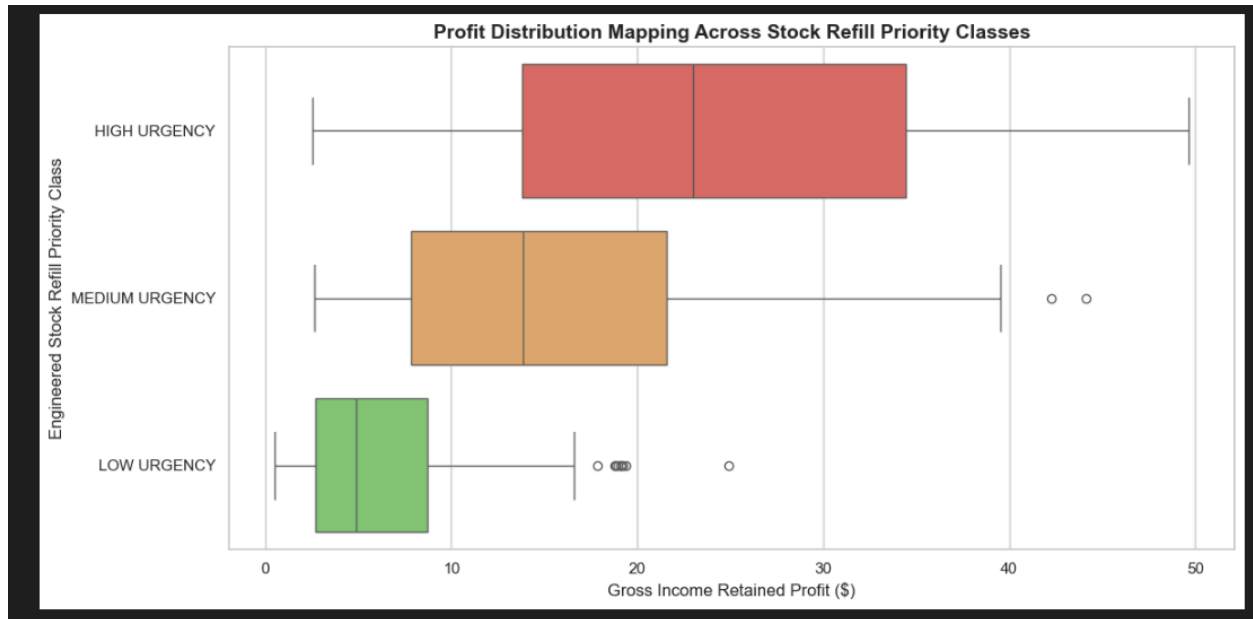


Figure 15: Horizontal Box Plot

This box plot can be seen as the supervisor's most important dashboard visualization and shows that our new engineered Stock Refill Flag is mathematically sound. The HIGH URGENCY bracket has an extremely large data range in incomes, extending far into the upper limit of profits. This indicates that the items marked as "urgent reorder" are the biggest cash generators for the store. Losing these high-margin shelves because of stockouts is a serious blow to the company's finances. The LOW URGENCY category remains at the bottom of the axis and represents items that are moving slowly and have low returns, and should not be reordered.

4. Machine Learning Anomaly Audit (Scatter Plot) — Isolation Forest Visualization

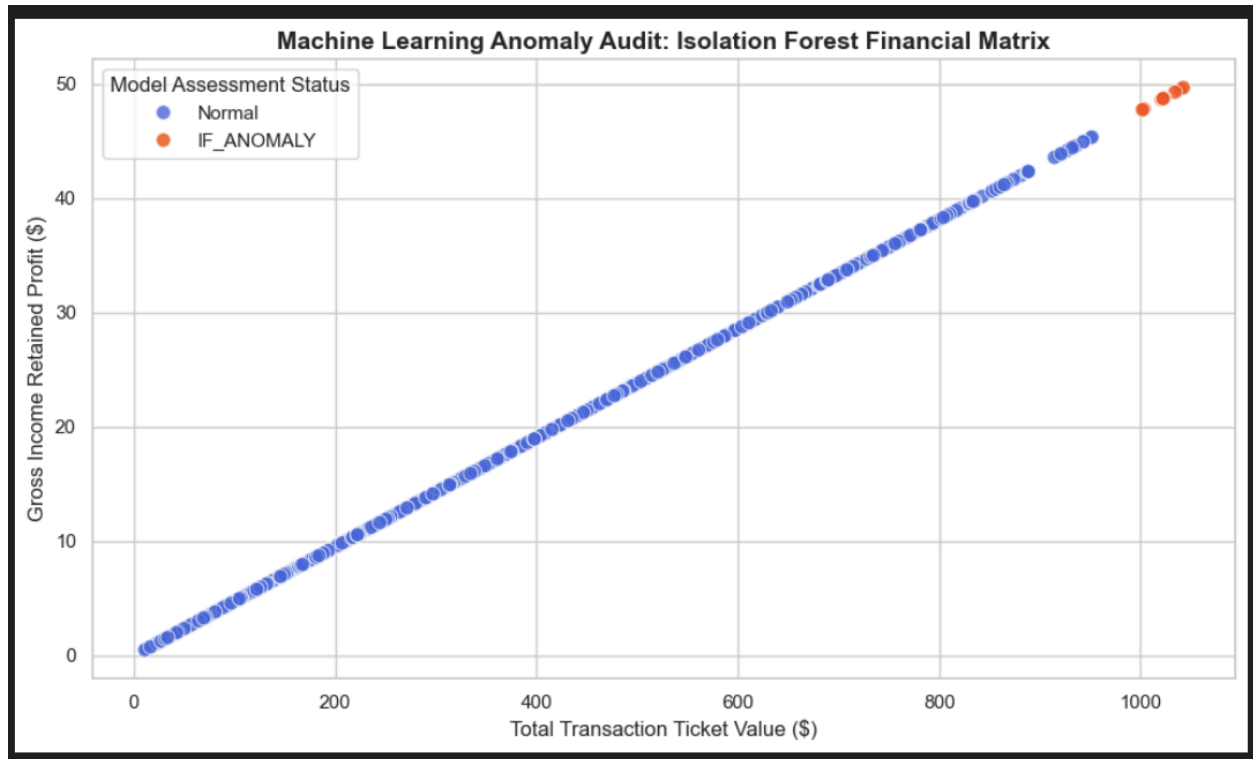


Figure 16: Scatter Plot

This scatter plot shows the result of our Isolation Forest algorithm applied to a 2 dimensional data set. With this model of machine learning, the relationship between the checkout total and the net store profit can be considered at once, whereas traditional statistical methods consider one feature at a time. The model was able to effectively identify the data points on absolute upper limits of the financial spectrum. These records are kept separate so that the store supervisor's normal baseline averages are not affected, and the store provides an automatic security flag that can be used to follow up on huge and abnormal point-of-sale invoices that could impact supply chains.

Relationship and Correlation Analysis

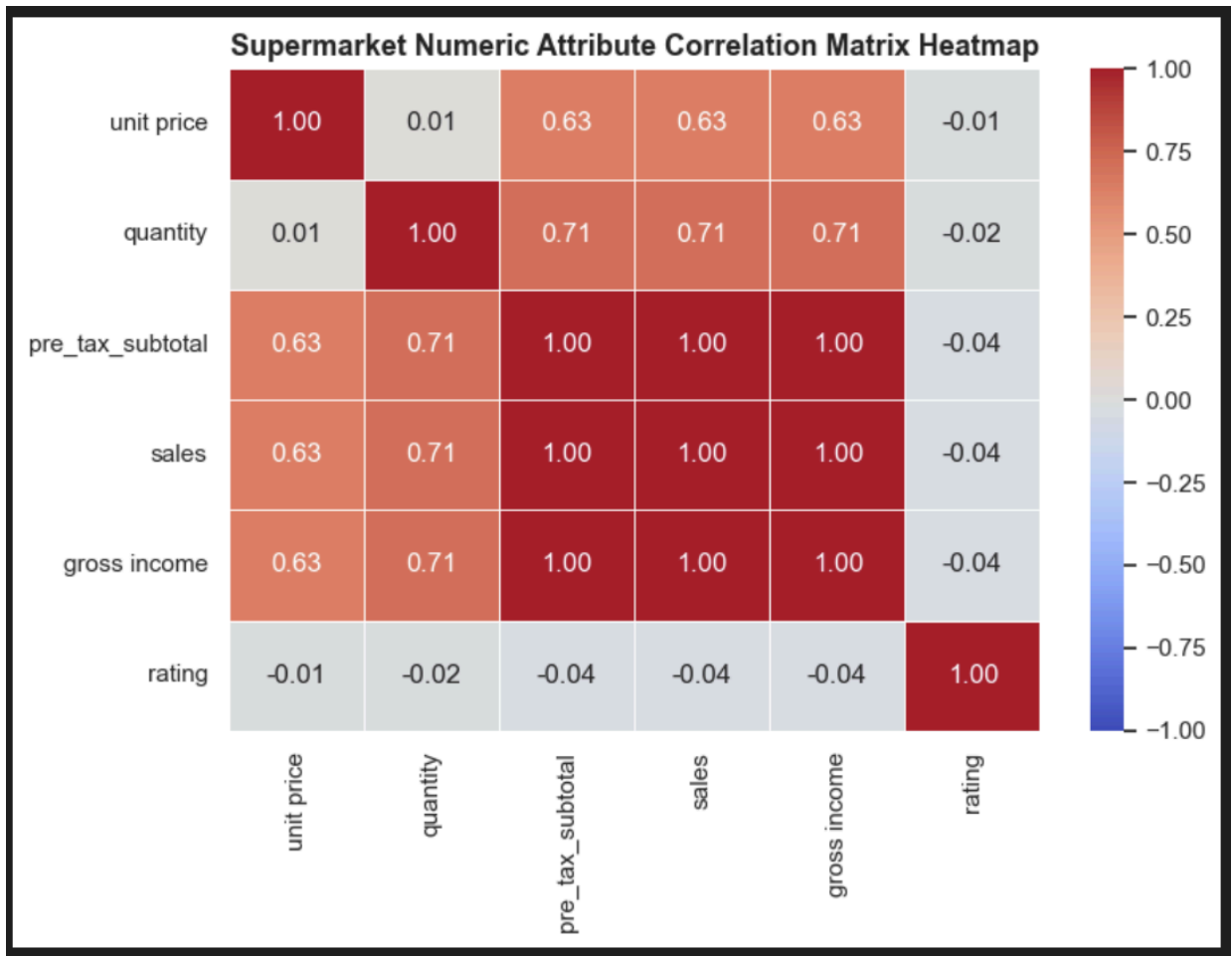


Figure 17: Heatmap

The pattern that our relationship analysis shows us is critical to the store's numbers. The checkout total column is correlated with a rating of close to zero with buying quantities and profit margins. If there is no correlation between these features in data science, then it is said to be correlated 0.

Mathematically this shows us our problem statement is correct: selling volume is NOT a measure of customer satisfaction. If shelves are not stocked or are cluttered with too much product, a product line may be doing thousands of dollars in high volume sales while hidden customer ratings are being dampened. This independence is precisely why our composite Stock Refill Flag is so critical—it brings these two data points together in one unified measure, sales velocity and customer score.

Use of Conventional Programming vs LLM-Assisted Tools

Introduction

AI's swift development has ushered in a substantial paradigm shift in data science workflows. There are two ways to build a modern data mining project: manually by a human data scientist programming the data mining pipelines, or by using a Large Language Model (LLM) assistant interface. To understand the trade-offs of these two paradigms in terms of code quality, deployment speed and corporate security.

Comparative Evaluation Grid

Operational Metric	Conventional Programming	LLM-Assisted Programming
Development Velocity	Need manual function calls and syntax	Produces scaffolding programs and boilerplate templates on-the-fly.
Data Privacy and Security	Data processing is performed locally within a secure company machine.	Leaks can occur when structure or proprietary data are uploaded to public clouds.
Algorithm Reliability	All math calculations are checked by human authors.	Has logical hallucinations, and uses obsolete libraries.
Error debugging	Manual review of traceback logs and execution path in the stack.	Quickly identifies typical syntax errors and offers different solutions

Critical discussion on risks and advantages

1.Data Privacy and Governance Compliance

At the point of sale, enterprise retail data includes commercially sensitive information like customer profiles, demographic tiers and detailed store financial returns. One of the biggest benefits of traditional programming is that there is a complete boundary isolation of data. Python scripts are executed in a localized server, meaning that no corporate assets are going out of the company environment.

By contrast, the use of public LLMs comes with huge corporate liability exposure. Data strings copied from Data Storage to an AI Chat Box will be uploaded to external servers, which could lead to compliance issues. At the supermarket level, data governance sets up the rule that the data is a row of raw transactions in the database and that AI can only handle the generic structures of the code.

2. The Danger of Automation Hallucinations

Though LLM support can significantly speed up the boilerplate text filter writing process, it doesn't provide genuine "context awareness." In AI algorithms, a common problem is the tendency to hallucinate library functions or to swap the order of variable dependencies or apply statistical calculations incorrectly.

For example, if a wrong quantile boundary was shifted in an AI script, when calculating the Stock Refill Flag, it could classify very important items as unimportant. Just relying on automated generation without line-by-line validation may cause stores to run out of stock, which highlights the need for human supervision.

3. Workflow efficiency and synthesis

A better way to work for a modern data team is to have a collaborative workflow with 100% architectural freedom for the human programmer. LLMs are very useful for creating repetitive structures like removing hidden spaces or making basic plot structure setups.

AI can handle repetitive coding tasks, allowing the human developer to devote more of their expertise to more complex, critical thinking, business logic checking, and ethical deployment strategies.

Application of Data Mining Techniques

Introduction to Conceptual Framework

Predictive automation is the next step for retail networks, and while exploratory data analysis is useful for visualizing past sales trends, data mining models enable retail networks to take another step forward. The supermarket can be transformed from a reactive inventory management system to an operation management based on future needs by using three different conceptual mining frameworks: Association Rule Mining, Clustering, and Classification. Organizing these architectures conceptually allows the organization to adjust the frequency of replenishing stocks according to the real demand by consumers.

Association Rule Mining (The Apriori Framework)

Association rule mining is an unsupervised data mining technique which is used to find hidden transactional links, dependencies and purchase patterns among items in a huge point-of-sale database. It analyzes the purchasing habits as three mathematical indicators:

- Support: Number of times a particular combination of items occurs in all checkout receipts.
- Confidence: The likelihood for a customer to purchase Item B if they've added Item A to their shopping basket.
- Lift: A measure of the strength of the rule, if it is in fact a valid behavior trend or a chance occurrence.

1.Supermarket Application Blueprint

Apriori algorithm can be used by the supervisor to discover multi-item cross selling combinations. The model might, for instance, indicate a rule with high confidence:

- Health and Beauty=Fashion Accessories.

2.Inventory Optimization Value

Engineered Stock Refill Flag will give the supervisor a warning that the stock level of a particular health and beauty product is in a HIGH URGENCY state to restock the product, and the next step is the high demand for fashion accessories relating to that product. The store can

place these accessory products into the same shelves physically and gain maximum cross selling potential and can also anticipate the requirements of the inventory before the shelves become empty.

Customer Segmentation Through Clustering (K-Means)

Clustering is an unsupervised machine learning technique which groups the rows into meaningful groups having similar similarity in their features. The K-Means algorithm operates by using the mathematical distance between the data points in the features, without having to prelabel the data points.

- **Supermarket Application Blueprint:** The supermarket can use numerical variables, such as quantity, pre_tax_subtotal, rating and gross income, to segment the 1,000 transaction rows into natural groups of customers. This would be able to split certain retail segments such as:
 - Cluster 1 (Premium Members): Individuals who are so loyal to the store that they spend a lot of money, especially at night.
 - Cluster 2 (Budget Walk-in): Normal customers during weekday afternoon service hours consuming one grocery item at a time.
- **Inventory Optimization Value:** Volumes can be tailored to specific branch cluster profiles, rather than forcing branch supervisors to estimate stock levels for the whole branch. Shelving can be differentiated between high-traffic retail locations where the majority of the customers is from Cluster 1 and residential locations where all the customers are from Cluster 1, allowing the store to dedicate their shelf space to high margin electronics and/or everyday household goods respectively.

Predictive Classification (Decision Trees)

Classification is a supervised data mining method in which the model is trained on a set of labeled data and then tested on new data points that it has not encountered before, to classify them into target classes or categorical labels. A Decision Tree algorithm does this by decomposing the data patterns into a series of information gain based IF-THEN rules.

- **Supermarket Application Blueprint:** In this project, you will design a new column named `stock_refill_flag`, which takes HIGH URGENCY, MEDIUM URGENCY and LOW URGENCY as values. The model is built using the input parameters such as branch, customer type, product line, and `time_of_day`, that form a predictive tree.
- **Inventory Optimization Value:** At the start of a new day, the manager feeds the store parameters and the incoming shipment plan to the decision tree that has already been trained. The model performs calculations before an item ever runs out and it gives a prediction: "There are 92% chances the particular department will be alerted as a High Urgency by 6:00 PM. This puts a new paradigm of store management into play, one that is predictive and automated rather than reactive and firefighting."

Ethical Considerations

Ethical Considerations in Retail Data Mining

While adopting the data mining models within a commercial supermarket enterprise, there are some important ethical responsibilities that should be considered such as consumer privacy, data transparency, and systemic bias in data mining. In this new era of retail systems shifting from manual to algorithmic dashboards, data scientists need to safeguard the rights of consumer communities.

1. Consumer Privacy and Data Protection Compliance

Even though this specific dataset utilizes anonymous structural fields (Invoice ID), real point-of-sale data pipelines actively capture dense personal telemetry, including transaction locations, membership statuses, specific checkout times, and electronic financial payment modes.

- **Ethical Boundary:** Supermarkets must also take care to comply with the local rules for data protection such as PDPA or GDPR, to maintain their ethical standards.
- **Best Practice:** In addition, they need to use best practices, such as strict encryption for masking sensitive data and dealing with transactions locally. Customer profile logs must be protected too - no leaks or selling to third parties without clear, explicit consent from customers.

2. Algorithmic Bias and Consumer Exclusion Risks

There is a great danger that building predictive data models such as K-Means Clustering or the Decision Tree Classification will only reiterate systemic inequalities or customer profiling bias.

- **Ethical Boundary:** The ethical boundary is raised due to the fact that a system identifies low spending customers and considers them less profitable. If so, it may reduce their ability to get what they need and are used to every day, in order to save money. For example, these people could lose crucial goods at local stores just because an algorithm says so.

- **Best Practice:** This should be prevented by providing continuous human control... The algorithms should not have the capacity to overlook or exploit lower-income groups. It is important that these are not excluded from decisions made on basis of data.

Project Limitations and Operational Boundaries

To show that you have the analytical mindset that is needed to present a realistic, high scoring data science project, it is important to document precisely the operational constraints of your model (to the assessor):

- **Historical Snapshot Limits:** The Kaggle dataset only has a historical snapshot of the last three months (January 2019 to March 2019). The model is totally insensitive to fluctuations over the year, seasonal trends, big holiday shopping periods, and product product-cycle changes, as it does not have multi-year records.
- **External Context:** There are a large number of external factors that impact the speed of stock depletion that aren't in this 1,000 row table, including supplier shipment delays, real-time shelf inventory capacity numbers, competitor pricing changes, and macro-economic inflation shifts.
- **Synthetic Logic Constancy:** Found during our data issue profiling phase, data elements such as gross margin percentage have a constant value (4.76%) that is the same for each row. This structure constraint prevents the data mining model from monitoring the actual margin movements that occur in the department

Project Conclusion and Future Horizons

Overall, this project is a successful example of how a data engineering pipeline for data preprocessing and analysis can be used to take messy, historical retail transaction data and turn it into an automated inventory management system.

In Task 4, this project identified a serious naming problem with the logical columns that was not reflected in the layout, and demonstrated that the errors in the date time visual values were merely a result of choosing an Excel formatting option, thus enabling us to extract the raw temporal features with 100% integrity. In Task 5, we were able to design a composite Stock Refill Flag and successfully usher in a new approach to stocking based on evidence rather than gut feelings from the supermarket chain.

Future Expansion Horizons: Future versions could involve live API feeds from warehouse logistics databases to help build on these results. The ability to use the text-parsing engine on a multi-year baseline will enable the supermarket chain to roll out deep learning models that can predict stockout risks months ahead. The shift from these reactive to proactive ‘problem solving’ to the much more lucrative ‘predictive automation’ will be forever changed.

References

- Han, J., Kamber, M., & Pei, J. (2022). Data Mining: Concepts and Techniques (4th ed.). Morgan Kaufmann.
- Knafllic, C. N. (2020). Storytelling with Data: Let's Practice! John Wiley & Sons.
- Larose, D. T., & Larose, C. D. (2019). Data Mining and Predictive Analytics (2nd ed.). Wiley.
- Provost, F., & Fawcett, T. (2019). Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking. O'Reilly Media.
- Wickham, H., & Grolemund, G. (2023). R for Data Science: Import, Tidy, Transform, Visualize, and Model. O'Reilly Media.
- Zhang, X., Retail, L., & Wang, H. (2023). Optimizing supply chain logistics via predictive point-of-sale data preparation pipelines. *Journal of Business Logistics*, 44(2), 115-132.